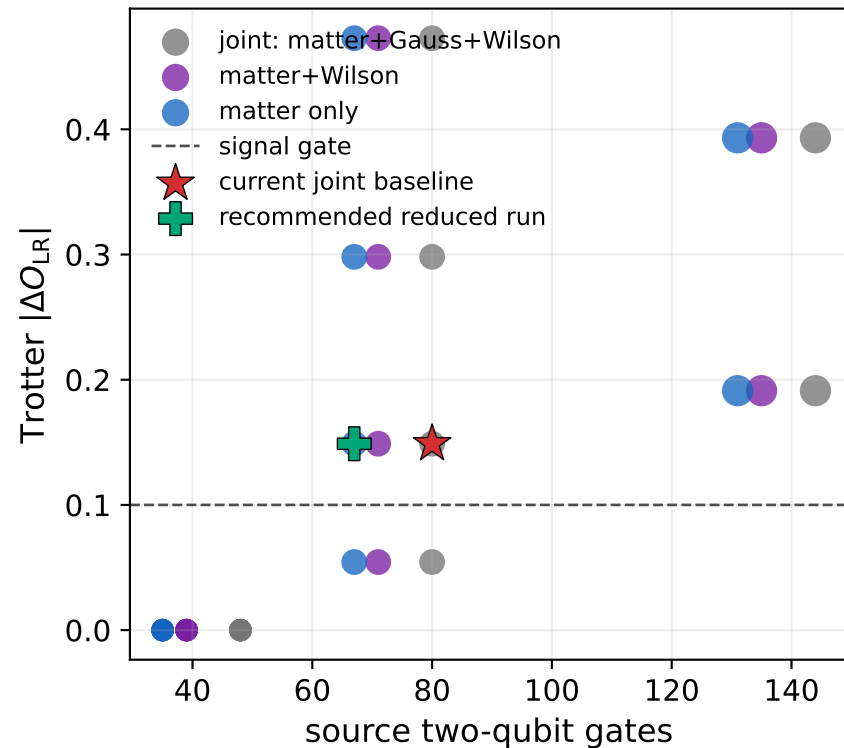
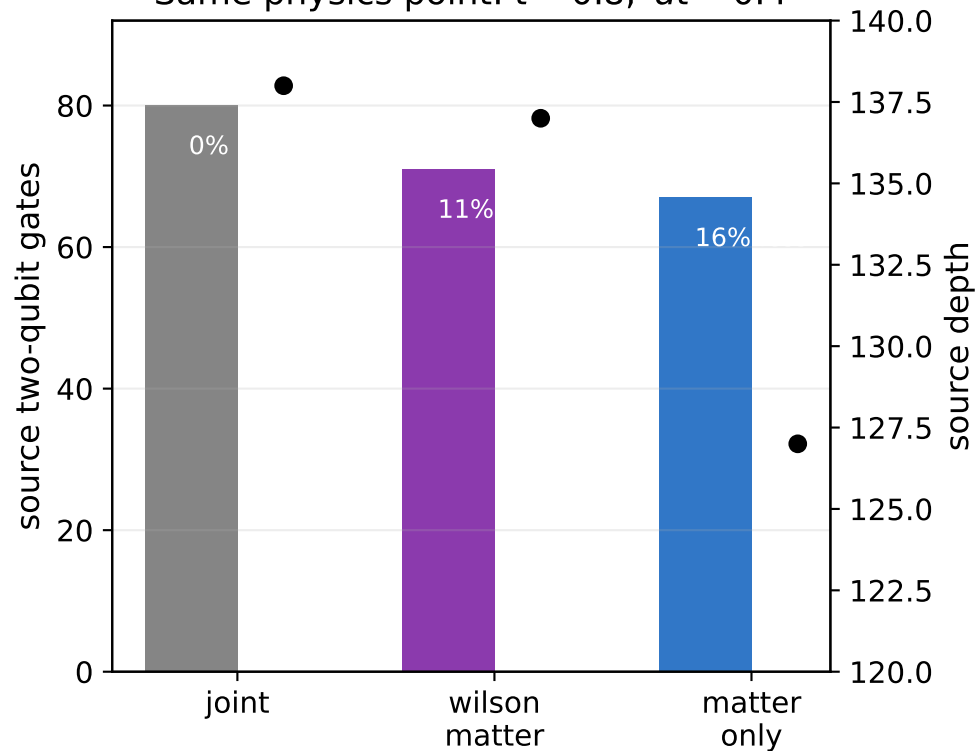


# Periodic hardware depth-reduction audit: split dynamics readout from diagnostics

## Signal versus source two-qubit count



## Same physics point: $t = 0.8$ , $dt = 0.4$



One-step Trotter candidates have zero target  $O_{LR}$  separation; two steps remain the minimum for this observable.